Sequences & Patterns

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Visual and growing patterns**

The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.

The rule is (pattern number)^2. Pattern 6 uses $6^2 = 36$ squares.

2 **[Foundational]** • **Target: Arithmetic sequences**

Find the 10th term of the sequence $5, 8, 11, 14, \ldots$

$a = 5$, $d = 3$. $a_{10} = 5 + 9 \times 3 = 32$.

3 **[Foundational]** • **Target: Arithmetic sequences**

Find the 15th term of the sequence $7, 12, 17, 22, \ldots$

$a = 7$, $d = 5$. $a_{15} = 7 + 14 \times 5 = 77$.

4 **[Foundational]** • **Target: Arithmetic sequences**

An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.

$a_{20} = 4 + 19 \times 6 = 4 + 114 = 118$.

5 **[Foundational]** • **Target: Arithmetic sequences**

The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.

$3 + 4d = 23 \Rightarrow 4d = 20 \Rightarrow d = 5$.

6 **[Foundational]** • **Target: Arithmetic sequences**

An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?

$7 + (n - 1) \times 5 = 82 \Rightarrow n - 1 = 15 \Rightarrow n = 16$. The 16th term.

7 **[Foundational]** • **Target: Arithmetic sequences**

The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.

$a + 7 \times 4 = 34 \Rightarrow a + 28 = 34 \Rightarrow a = 6$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Geometric sequences

Find the next term of the sequence $2, 4, 8, 16, \ldots$

Geometric, ratio 2: $16 \times 2 = 32$.

9 [Intermediate] • Target: Geometric sequences

Find the 6th term of the sequence $3, 6, 12, 24, \ldots$

$a = 3$, $r = 2$. $a_6 = 3 \times 2^5 = 3 \times 32 = 96$.

10 [Intermediate] • Target: Geometric sequences

A geometric sequence has first term 5 and common ratio 3. Find the 5th term.

$a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

11 [Intermediate] • Target: Geometric sequences

The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.

$2 \times r^3 = 54 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

12 [Intermediate] • Target: Geometric sequences

A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.

$r = 18 \div 6 = 3$. First term $= 6 \div 3 = 2$.

13 [Intermediate] • Target: Identifying the type of sequence

Is $5, 10, 20, 40, \ldots$ arithmetic, geometric, or neither?

Each term is double the previous one ($r = 2$): geometric.

14 [Intermediate] • Target: Identifying the type of sequence

Is $6, 10, 14, 18, \ldots$ arithmetic, geometric, or neither?

Each term is 4 more than the previous one ($d = 4$): arithmetic.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identifying the type of sequence • Cross-Lesson: 5.01 Quantitative Comparisons

Is $1, 4, 9, 16, \ldots$ arithmetic, geometric, or neither?

Differences $3, 5, 7$ are not constant, and ratios are not constant either: neither (this is the sequence of perfect squares, n^2).

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16 [Challenging] • Target: Identifying the type of sequence • Cross-Lesson: 7.01 Data Analysis & Probability

Is $100, 90, 81, 72.9, \ldots$ arithmetic, geometric, or neither?

Each ratio is $\frac{90}{100} = \frac{81}{90} = \frac{72.9}{81} = 0.9$: geometric with $r = 0.9$.

17 [Challenging] • Target: Finding missing terms • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the missing term: $3,\ __,\ 11,\ 15$ (arithmetic).

Common difference $= (15 - 3) \div 3 = 4$. Missing term $= 3 + 4 = 7$.

18 [Challenging] • Target: Finding missing terms • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the missing term: $4,\ __,\ 36$ (geometric, all terms positive).

$r = \sqrt{36 \div 4} = \sqrt{9} = 3$. Missing term $= 4 \times 3 = 12$.

19 [Challenging] • Target: Finding missing terms • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Insert two arithmetic means between 5 and 20, forming the four-term sequence $5, __, __, 20$.

Common difference $= (20 - 5) \div 3 = 5$. The sequence is $5, 10, 15, 20$: the missing terms are 10 and 15.

20 [Challenging] • Target: Finding missing terms • Cross-Lesson: 3.01 Algebra Essentials

The sequence $x,\ 2x + 1,\ 4x - 1$ is arithmetic. Find x.

Equal consecutive differences: $(2x + 1) - x = (4x - 1) - (2x + 1) \Rightarrow x + 1 = 2x - 2 \Rightarrow x = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.

a. 37

b. 40

c. 29

d. 33 ✓

$d = 4$, $a_1 = 5$: $T_8 = 5 + 7(4) = 33$.

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22 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.

a. 34

b. 43

c. 40

d. 37 ✓

$d = 3$, $a_1 = 4$: $T_{12} = 4 + 11(3) = 37$.

23 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.

a. 41

b. 51

c. 50

d. 46 ✓

$d = 5$, $a_1 = 6$: $T_9 = 6 + 8(5) = 46$.

24 [Foundational] • Target: MCQ — Arithmetic sequences

Find the 12th term of the sequence $4, 9, 14, 19, \ldots$

a. 49

b. 54

c. 64

d. 59 ✓

$a_1 = 4$, $d = 5$: $T_{12} = 4 + 11(5) = 59$.

25 [Foundational] • Target: MCQ — Arithmetic sequences

An arithmetic sequence has first term 10 and common difference 6. Which term equals 100?

a. 17th

b. 90th

c. 15th

d. 16**th** ✓

$100 = 10 + (n-1)(6) \Rightarrow n - 1 = 15 \Rightarrow n = 16$.

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26 [Foundational] • Target: MCQ — Geometric sequences

A geometric sequence has first term 4 and common ratio 2. Find the 6th term.

a. 256

b. 32

c. 128 ✓

d. 64

$T_6 = 4(2^5) = 128$.

27 [Foundational] • Target: MCQ — Identifying the type of sequence

Is $7, 14, 28, 56, \ldots$ arithmetic, geometric, or neither?

a. Neither

b. Arithmetic

c. **Geometric** ✓

d. Both arithmetic and geometric

Each term is double the last (ratio 2): geometric.

28 [Foundational] • Target: MCQ — Finding missing terms

Find the missing term: $5,\ __,\ 80$ (geometric, all terms positive).

a. 42.5

b. 20 ✓

c. 16

d. 40

$r^2 = 80 \div 5 = 16 \Rightarrow r = 4$; missing $= 5 \times 4 = 20$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Finding missing terms

Find the missing term: $__,\ 12,\ 19,\ 26$ (arithmetic).

a. 7

b. 9

c. 4

d. 5 ✓

$d = 7$: missing $= 12 - 7 = 5$.

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30 [Intermediate] • Target: MCQ — Sum of an arithmetic series

An arithmetic sequence has first term 5 and common difference 3. Find the sum of its first 20 terms.

a. 685 **b.** 670 ✓

c. 640 **d.** 625

$S_{20} = \frac{20}{2}(2(5) + 19(3)) = 10(67) = 670$.

31 [Intermediate] • Target: MCQ — Sum of a geometric series

Find the sum of the first 4 terms of the sequence $1, 3, 9, 27, \ldots$

a. 40 ✓ **b.** 81

c. 27 **d.** 39

$1 + 3 + 9 + 27 = 40$.

32 [Intermediate] • Target: MCQ — Special sequences

A sequence starts $3, 5, 8, 13, 21, \ldots$, where each term is the sum of the two terms before it. Find the 8th term.

a. 55 **b.** 89 ✓

c. 144 **d.** 76

Continuing: $3, 5, 8, 13, 21, 34, 55, 89$: the 8th term is 89.

33 [Intermediate] • Target: MCQ — Word problems

A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

a. 60 **b.** 15

c. 3.75 **d.** 7.5 ✓

$a_1 = 120$, $r = 0.5$: $T_5 = 120(0.5)^4 = 7.5$ cm.

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34 [Intermediate] • Target: MCQ — Pattern reasoning

A sequence follows the rule \"triple and subtract 2\", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82 ✓

d. 250

$4 \to 10 \to 28 \to 82$: the 4th term is 82.

35 [Intermediate] • Target: MCQ — Sum of an arithmetic series

Find the 13th term of the sequence $-14,\ -21,\ -28,\ -35, \ldots$

a. -98 ✓

b. -105

c. -96

d. -91

$a = -14, d = -7$. $a_{13} = -14 + 12 \times (-7) = -98$.

36 [Intermediate] • Target: MCQ — Sum of a geometric series

Find the 14th term of the sequence $12, 21, 30, 39, \ldots$

a. 131

b. 138

c. 129 ✓

d. 120

$a = 12, d = 9$. $a_{14} = 12 + 13 \times 9 = 129$.

37 [Intermediate] • Target: MCQ — Special sequences

Find the 6th term of the sequence $-2, 1, 4, 7, \ldots$

a. 10

b. 13 ✓

c. 16

d. 15

$a = -2, d = 3$. $a_6 = -2 + 5 \times 3 = 13$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Visual and growing patterns • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the 14th term of the sequence $-16, -22, -28, -34, \ldots$

a. -94 ✓

b. -100

c. -92

d. -88

$a = -16, d = -6$. $a_{14} = -16 + 13 \times (-6) = -94$.

39 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the 7th term of the sequence $-1, 3, -9, 27, \ldots$

a. -729 ✓

b. 729

c. -727

d. -731

$a = -1, r = -3$. $a_7 = -1 \times (-3)^6 = -1 \times 729 = -729$.

40 [Challenging] • Target: MCQ — Pattern reasoning • Cross-Lesson: 3.01 Algebra Essentials

Find the 6th term of the sequence $6, 12, 24, 48, \ldots$

a. 194

b. 190

c. 192 ✓

d. -192

$a = 6, r = 2$. $a_6 = 6 \times 2^5 = 6 \times 32 = 192$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Geometry & Measurement

Find the 3rd term of the sequence $-5, 15, -45, 135, \ldots$

a. -43

b. 45

c. -47

d. -45 ✓

$a = -5, r = -3$. $a_3 = -5 \times (-3)^2 = -5 \times 9 = -45$.

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42 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Quantitative Comparisons

Find the sum of the first 14 terms of the sequence $-3, -11, -19, -27, \ldots$

a. -766

b. -774

c. -778

d. -770 ✓

$a = -3, d = -8, n = 14. S = \frac{n}{2}(2a + (n-1)d) = 7(-6 - 104) = 7 \times (-110) = -770.$

43 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Data Analysis & Probability

Find the sum of the first 6 terms of the sequence $-4, -8, -16, -32, \ldots$

a. -255

b. -249

c. -256

d. -252 ✓

$S_6 = \frac{a(2^6 - 1)}{r - 1} = \frac{-4(63)}{1} = -252.$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the sum of the first 5 terms of the sequence $9, -27, 81, -243, \ldots$

a. 546

b. 558

c. 552

d. 549 ✓

$S_5 = \frac{a((-3)^5 - 1)}{r - 1} = \frac{9(-244)}{-4} = 549.$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the missing term: $1, __, 9$ (geometric, all terms positive).

a. 3 ✓

b. 6

c. 10

d. 1

$r = \sqrt{9 \div 1} = \sqrt{9} = 3$. Missing term $= 1 \times 3 = 3.$

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